The UPI Impulse Trap: Behavioral Analysis of Impulsive Digital Spending Patterns and Financial Regret Among Indian College Students

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Abstract

The rapid adoption of Unified Payments Interface (UPI) among Indian college students has introduced frictionless digital spending that may facilitate impulsive purchasing behavior. This study investigates the behavioral and psychological triggers behind UPI-based impulse spending and subsequent financial regret among 105 college students across four institutions in India. Using a structured survey instrument spanning six sections, we collected primary data covering demographics, spending behavior, impulse trigger scores, regret patterns, and open-ended regret descriptions. We applied Exploratory Data Analysis (EDA), Natural Language Processing (NLP) using TextBlob sentiment analysis and word frequency analysis, K-Means clustering to identify three distinct spending archetypes, and a Random Forest classifier to predict high financial regret risk. Results reveal that 69.5% of respondents make unplanned purchases in the evening or late-night hours, food delivery is the most regretted category (48.6% of respondents), stress relief and exam stress are the dominant impulse triggers, and 33.3% of the sample falls into a high financial regret group. The impulse composite score and weekly transaction frequency are the strongest predictors of financial regret, as identified by feature importance analysis. Three behavioral clusters are identified: Routine Evening Spenders (62.9%), Controlled Spenders (29.5%), and High Impulsive Spenders (7.6%).

Keywords

UPI, impulse buying, financial regret, college students, K-Means clustering, Random Forest, NLP, behavioral analytics, India

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1 Introduction

India crossed 500 million Unified Payments Interface (UPI) users in 2024, with college students among the most active adopters. UPI enables near-instant digital payments through mobile applications such as Google Pay, PhonePe, and Paytm, removing traditional friction associated with physical cash or card transactions. Behavioral economists describe the "pain of paying" phenomenon [?]: the more abstract and invisible a payment method, the lower the psychological resistance to spending. UPI represents the most frictionless payment mechanism available in the Indian context, with no physical card, no visible cash depletion, and no PIN delay in many configurations.

This frictionlessness has a behavioral consequence. College students operating on limited monthly budgets, often funded entirely by parents, are exposed to an environment where digital spending requires minimal deliberation. Food delivery platforms, quick-commerce applications, and online marketplaces deliver gratification within minutes of a UPI transaction. The result is a pattern of unplanned purchases followed by post-purchase regret that is understudied in the Indian college context.

This project makes the following contributions:

- A primary dataset of 105 college students across four Indian institutions covering spending behavior, impulse triggers, and financial regret, collected via a structured Google Form survey.
- Exploratory analysis revealing that evening and late-night hours account for the majority of impulse purchases, and food delivery is the dominant regret category.
- NLP analysis of open-ended regret descriptions using TextBlob sentiment scoring and word frequency extraction.
- K-Means clustering identifying three distinct spending behavioral archetypes.
- A Random Forest classifier predicting high financial regret risk with identification of the most predictive behavioral features.

2 Related Work

2.1 Pain of Paying and Payment Abstraction

Prelec and Simester [?] experimentally demonstrated that payment method abstraction reduces the psychological cost of spending. Using credit cards versus cash, they showed that more abstract payment modes increase willingness to pay. UPI extends this abstraction further: no physical medium is involved, and the transaction completes in under three seconds. This theoretical framework directly motivates our hypothesis that UPI use is associated with elevated impulsive purchasing behavior.

2.2 Mobile Payments and Impulse Buying

Research on mobile payment platforms and impulse buying has grown since the proliferation of digital wallets. Studies in the mobile commerce literature identify convenience, reduced transaction time, and cashback rewards as key facilitators of unplanned purchases [?

DAV Major Project '26, IIT Bhilai, India
2026.

]. In the Indian context, UPI cashback incentives and limited-time offers from apps like Zomato and Blinkit create promotional triggers that encourage spending beyond planned budgets.

2.3 Student Financial Behavior

College students represent a financially vulnerable demographic: income is limited, financial literacy is often low, and social influences on spending are strong. Prior work has documented that students are susceptible to peer-influenced purchasing (FOMO), emotional spending during academic stress periods, and habitual late-night online consumption facilitated by mobile devices [**?**].

3 Methodology

3.1 Survey Design

A structured Google Form was designed with six sections:

- **Section A – Demographics:** Age group, gender, college year, monthly budget range, exact monthly budget, income source.
- **Section B – UPI App Usage:** Primary UPI application, perception of UPI as an overspending risk, and reasons for offline UPI use.
- **Section C – Spending Behavior:** Weekly transaction count (range and exact), percentage of unplanned transactions, time-of-day profile for impulse purchases, and balance-checking habits.
- **Section D – Financial Habits:** Expense tracking behavior, frequency of running out of money before month end, and hidden purchase behavior.
- **Section E – Impulse Triggers (Likert Scale):** Eight items rated 1 (Strongly Disagree) to 5 (Strongly Agree): boredom-driven purchasing, FOMO/social influence, late-night scrolling, cashback justification, stress relief buying, scarcity notifications, cart abandonment return, and exam-season stress spending.
- **Section F – Regret Mapping:** Monthly regret frequency, regret category (multi-select), post-purchase regret intensity (1–10 scale), post-regret coping action, and an open-ended description of a regretted purchase.

All humor-labeled answer choices (e.g., "I am scared of what I'll find") were preserved to encourage honest responses and later standardized during cleaning.

3.2 Data Collection

The survey was distributed across four college groups between March and April 2026:

- IIT Bhilai (54 responses)
- Jaipur Colleges (30 responses)
- AIIMS Jodhpur (12 responses)
- IIT Delhi (9 responses)

Distribution channels included WhatsApp broadcast, direct sharing in college networks, and the Reddit communities r/IndianTeens and r/personalfinanceindia. A total of **105 valid responses** were collected. Respondent email addresses were collected optionally and stored separately for report delivery, never used in analysis.

3.3 Data Cleaning and Preparation

All preprocessing was performed in Python using the Pandas library. The key steps were:

(1) **Respondent ID assignment:** Each response was assigned a unique numeric ID with a prefix indicating the source institution.
(2) **Joke label standardization:** All answer options containing appended humor text were stripped to retain only the categorical value (e.g., "Never (teach us your ways)" → "Never").
(3) **Ordinal encoding:** Categorical ordinal columns were mapped to integers: `balance_check_habit` (Never=0, Rarely=1, Sometimes=2, Always=3), `tracks_expenses` (No=0, Tries but quits=1, Yes=2), `ran_out_of_money` (Never=0, Rarely=1, Occasionally=2, Regularly=3), `hidden_purchase` (No=0, Maybe=1, Yes=2), `regret_frequency` (Never=0 through 10+=4).
(4) **Numeric midpoint imputation:** For range-type fields (monthly budget range, weekly transaction range, % unplanned), exact numeric values were used where provided; midpoints of the stated range were substituted otherwise.
(5) **Multi-select binary encoding:** The `impulse_time` and `regret_categories` multi-select fields were exploded into binary flag columns (e.g., `flag_latenight`, `cat_food_delivery`).
(6) **Outlier capping:** Four respondents reported exact monthly budgets exceeding Rs.10,000 within the Rs.6,000+ bracket; these were capped at Rs.7,000.
(7) **Target variable creation:** `high_regret` was defined as 1 if `regret_intensity` ≥ 6, else 0.
(8) **Impulse composite score:** The mean of all eight Likert trigger columns per respondent, producing a continuous behavioral index.

The final cleaned dataset contains 105 rows and 46 columns with only 23 null values, all in the optional open-ended text column.

3.4 Analytical Methods

3.4.1 Exploratory Data Analysis. Descriptive statistics and visualizations were produced using Matplotlib and Seaborn. Charts include demographic distributions, spending behavior frequency plots, correlation heatmaps, and gender-disaggregated trigger comparisons.

3.4.2 Natural Language Processing. TextBlob was used to compute polarity ($p \in [-1, 1]$, where -1 is most negative and $+1$ is most positive) and subjectivity ($s \in [0, 1]$, where 1 is maximally subjective) for each open-ended regret description. Word frequency analysis was performed after tokenization and stopword removal using NLTK. A word cloud was generated from the corpus using the `wordcloud` Python library.

3.4.3 K-Means Clustering. K-Means partitions n observations into k clusters by minimizing within-cluster sum of squares (WCSS):

$$\arg\min_{S} \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \tag{1}$$

where S_i is the set of points in cluster i and $\boldsymbol{\mu}_i$ is the cluster centroid. Features were standardized using Z-score normalization prior to clustering:

$$z = \frac{x - \mu}{\sigma} \tag{2}$$

The optimal number of clusters k was determined using the elbow method, plotting WCSS against k from 2 to 11 and identifying the inflection point. Principal Component Analysis (PCA) was applied to reduce the feature space to two dimensions for visualization.

3.4.4 Random Forest Classifier. A Random Forest is an ensemble of T decision trees, each trained on a bootstrap sample of the data with a random subset of features at each split. The final prediction is obtained by majority vote:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^{T} \tag{3}$$

where h_t is the prediction of the t-th tree. Feature importance is measured by the mean decrease in Gini impurity across all trees:

$$\text{Importance}(f) = \frac{1}{T} \sum_{t=1}^{T} \sum_{n \in N_t(f)} \Delta G(n) \tag{4}$$

where $N_t(f)$ is the set of nodes in tree t that split on feature f and $\Delta G(n)$ is the reduction in Gini impurity at node n. The model was trained with $T = 100$ trees on an 80/20 stratified train-test split.

4 Results

4.1 Demographic Profile

The sample comprised 105 respondents: 71 male (67.6%) and 34 female (32.4%). The majority (91, or 86.7%) were undergraduates; 14 (13.3%) were postgraduates. Age distribution was predominantly 18–21 years (78 respondents, 74.3%). All respondents were college students at Indian higher education institutions.

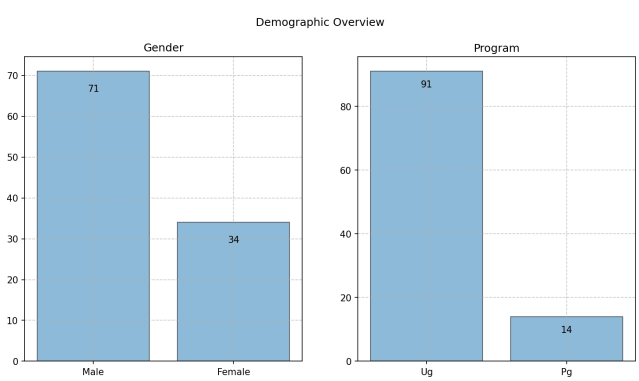

Figure 1: Demographic distribution: Gender (left) and Program (right).

The primary UPI application was Google Pay (GPay) used by 44.8% of respondents, followed by PhonePe (26.7%) and Paytm (14.3%). The three minor-use apps (Bhim UPI, Navi UPI, Other) each constituted 4.8%.

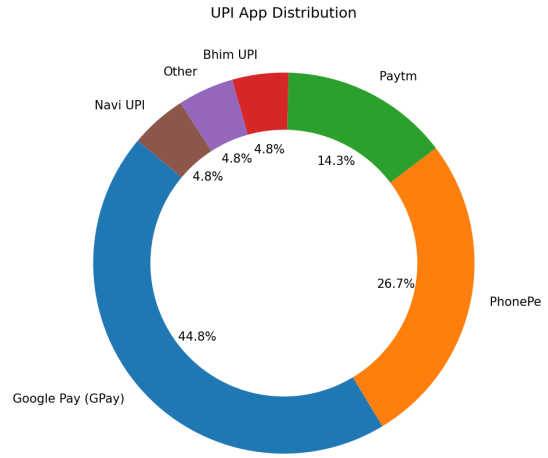

Figure 2: UPI application market share among respondents.

4.2 Spending Behavior

The largest budget segment was Rs.1,000–3,000 per month (50 respondents, 47.6%), consistent with a predominantly parent-funded student population.

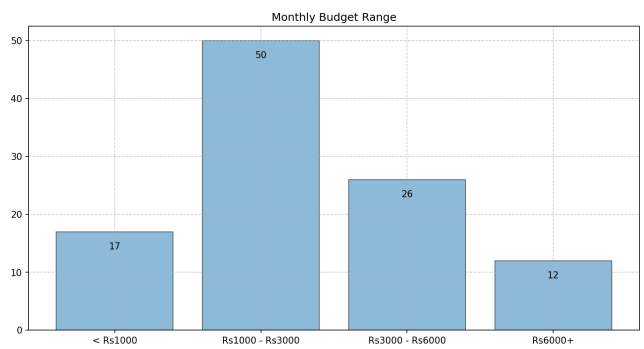

Figure 3: Monthly budget range distribution across respondents.

Transaction frequency was highest in the 4–7 per week bracket (40 respondents, 38.1%), with 19% making 15 or more transactions weekly. Regarding unplanned purchases, 37.1% reported 20–40% unplanned and 39 respondents (37.1%) reported 40–60% unplanned, indicating that the modal respondent makes roughly half their transactions without prior intent.

The timing of impulse purchases is strongly concentrated in the evening and late-night hours, as shown in Figure 5. Of 105 respondents, 73 (69.5%) reported making impulse purchases in the evening, and 42 (40%) during late night (10 PM–2 AM). Post-midnight purchases were reported by 9 respondents (8.6%).

4.3 Regret Patterns

Monthly regret frequency was low for most respondents, with 51 (48.6%) reporting 1–2 regretted purchases per month and 28 (26.7%)

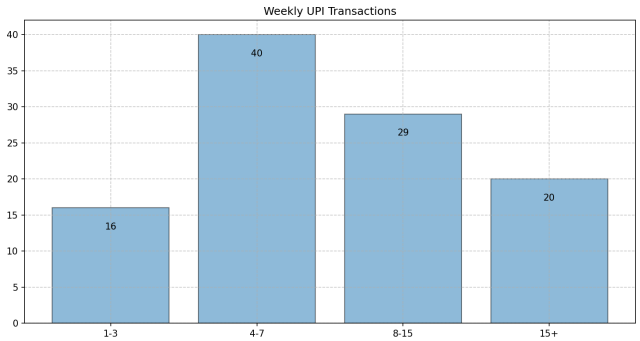

Figure 4: Weekly UPI transaction frequency distribution.

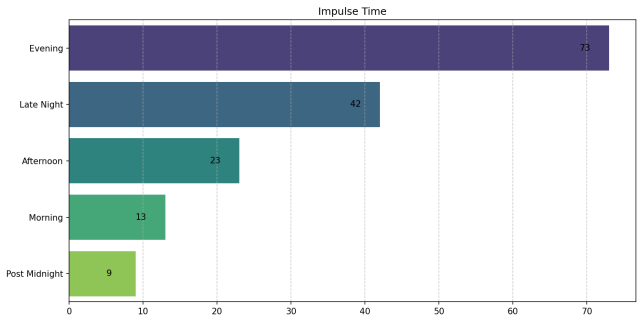

Figure 5: Time-of-day profile for impulse purchases (multi-select).

reporting none. However, 26 respondents (24.8%) reported 3 or more regretted purchases monthly.

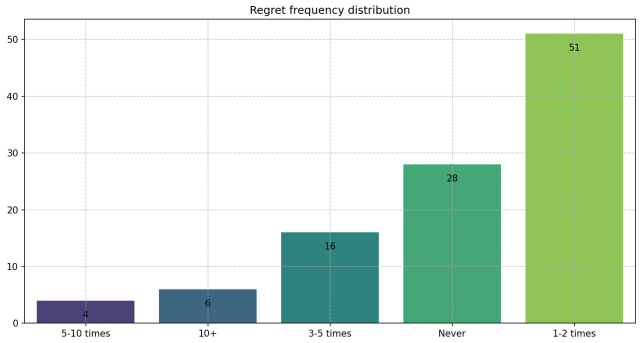

Figure 6: Monthly regret frequency distribution.

Food delivery (Swiggy/Zomato) was the most regretted category (51 respondents), closely followed by offline cafe and canteen spending (46). Grocery quick-commerce (Blinkit/Instamart) ranked third (33).

Regret intensity, rated 1–10, had a mean of 4.35 (SD $\approx$ 2.6). The distribution shows a concentration at intensity 3 (19 respondents) and a secondary cluster at intensity 7 (11 respondents). Using a threshold of 5.5, respondents were classified as high regret ($\geq$ 6, n = 35, 33.3%) or low regret ($\leq$ 5, n = 70, 66.7%).

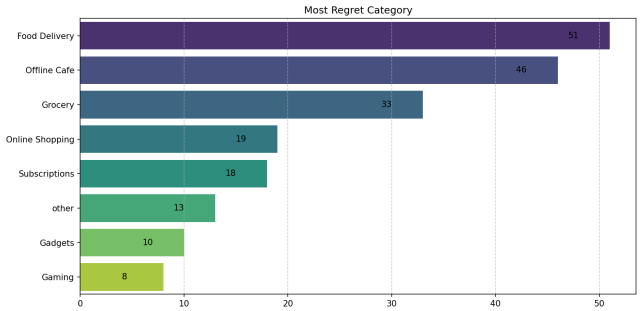

Figure 7: Regret category distribution (multi-select, sorted by frequency).

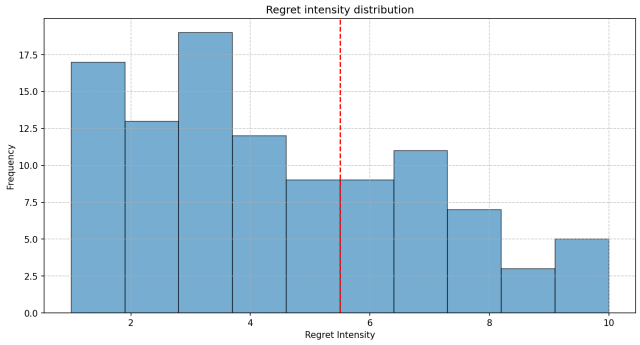

Figure 8: Regret intensity distribution (1–10) with high-regret threshold at 5.5.

4.4 Impulse Trigger Analysis

All eight Likert trigger items were scored 1–5. Mean scores are reported in Table 1.

Table 1: Mean Likert Scores for Impulse Trigger Items (1–5 scale)

Trigger	Mean Score	Max Observed
Stress Relief Buying	2.07	5
Exam Season Spending	2.06	5
Cashback Justification	1.91	5
FOMO / Social Influence	1.71	5
Late-Night Scrolling	1.70	5
Boredom-Driven Buying	1.57	5
Scarcity Notifications	1.53	4
Cart Abandonment Return	1.41	4

Stress relief and exam season spending ranked highest, indicating that academic pressure is a more significant impulse trigger than social influence or promotional mechanics for this sample. A gender-disaggregated comparison reveals that females score slightly higher on stress relief buying while males score higher on exam season spending.

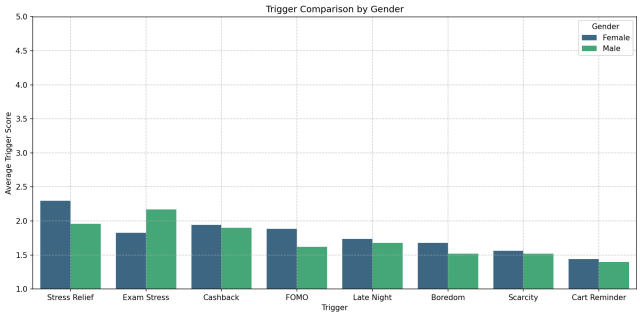

Figure 9: Average impulse trigger scores by gender (scale 1–5, y-axis starts at 1.0).

The mean impulse composite score (mean of all 8 triggers) across the full sample was 1.74 out of 5, indicating that on average respondents leaned toward disagreement with impulsive trigger statements. However, the distribution was right-skewed, meaning a minority of students showed substantially elevated composite scores.

The correlation heatmap (Figure 10) reveals the strongest positive correlation between regret frequency and regret intensity ($r = 0.38$), and between regret frequency and running out of money ($r = 0.33$). Balance checking habit shows a small negative correlation with unplanned purchasing ($r = -0.15$), consistent with the expectation that financial mindfulness reduces impulsive spending.

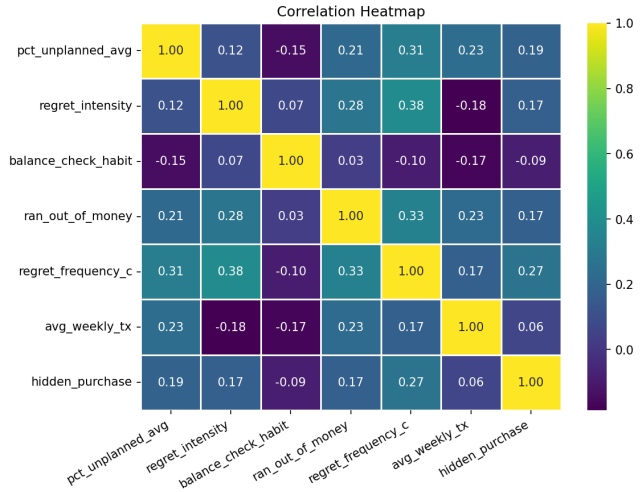

Figure 10: Pearson correlation heatmap of key numeric behavioral variables.

4.5 Hidden Purchase Behavior

42% of male respondents and 35% of female respondents reported having hidden a purchase from parents or roommates. This behavioral indicator captures implicit guilt associated with a purchase and was included as a feature in the predictive model.

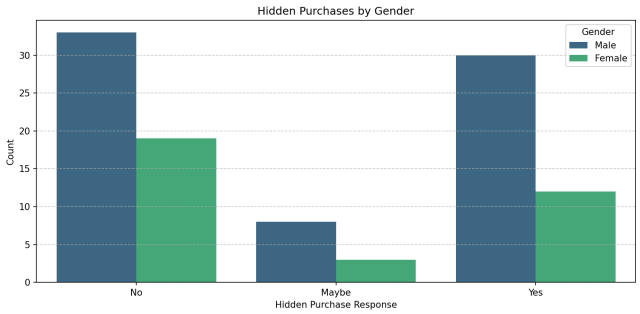

Figure 11: Hidden purchase behavior by gender (No, Maybe, Yes).

4.6 NLP Analysis

Of 105 respondents, 21 provided a response to the open-ended question: *"Describe a recent online/UPI purchase you regretted. What did you buy? What were you feeling right before you bought it? What made you click Pay?"*

TextBlob sentiment analysis yielded a mean polarity of 0.025 (near neutral with slight positive skew) and mean subjectivity of 0.333 (low-to-moderate subjectivity). The Pearson correlation between polarity score and self-reported regret intensity was $r = 0.020$, indicating that the linguistic sentiment of regret descriptions did not strongly predict quantitative regret intensity in this small sample.

Word frequency analysis (Figure 12) of the cleaned, stopword-removed corpus identified *food* as the highest-frequency term (9 occurrences), followed by *bought*, *feeling*, *money*, and *didnt*. The presence of *return*, *regret*, and *dont* confirms that descriptions were genuinely regret-oriented.

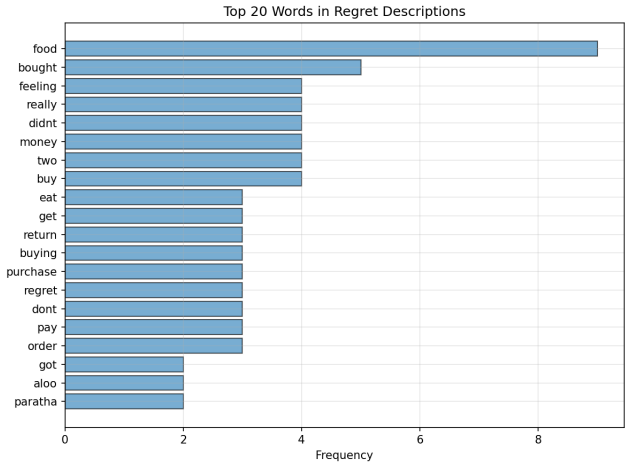

Figure 12: Top 20 words in open-ended regret descriptions after stopword removal.

Manual qualitative tagging of the 21 descriptions into trigger categories (Figure 14) identified that Emotional (19%), Poor Planning (19%), Cravings (19%), and Need (19%) were the four most common

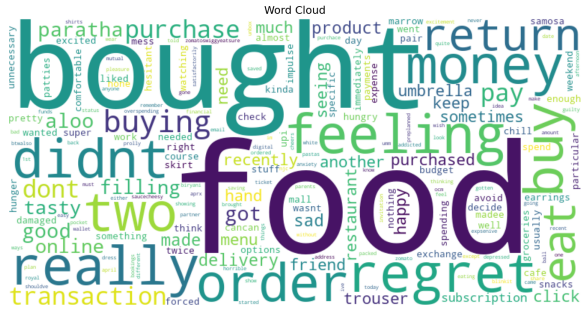
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Figure 16: PCA visualization of K-Means clusters. High Impulsive Spenders (orange) are clearly separated along PCA1.

Table 3: Random Forest Classification Report

Class	Precision	Recall	F1	Support
Low Regret (0)	0.69	0.79	0.73	14
High Regret (1)	0.40	0.29	0.33	7
Weighted Avg	0.59	0.62	0.60	21

Time-of-day flags contributed relatively less, though flag_latenight and flag_postmidnight were present in the High Impulsive cluster.

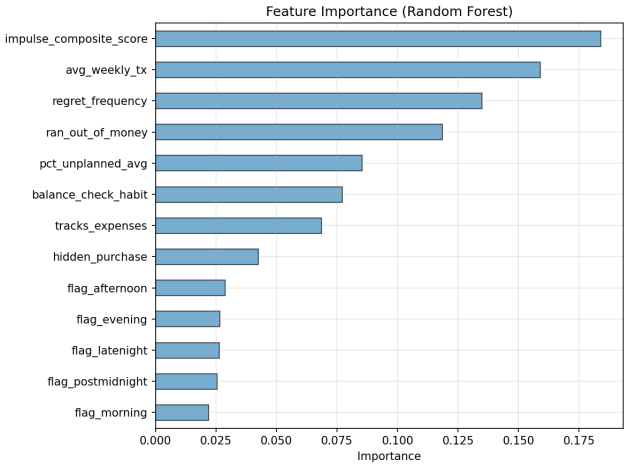

Figure 17: Random Forest feature importance scores, sorted descending.

5 Discussion

5.1 Evening and Late-Night Spending as the Primary Risk Window

The finding that 69.5% of respondents impulse-buy in the evening and 40% during late night is consistent with existing research on digital device usage patterns among college students. The absence of social oversight, combined with access to food delivery and e-commerce platforms that operate 24 hours, creates a structural environment for unplanned spending. The High Impulsive cluster shows 100% late-night and post-midnight purchasing, suggesting that nocturnal UPI activity is a reliable behavioral signature of elevated regret risk.

5.2 Stress and Academic Pressure as Dominant Triggers

Contrary to common assumptions that impulse buying is primarily driven by social influence (FOMO) or promotional mechanics (cashback, flash sales), the highest-scoring triggers were stress relief buying (mean 2.07) and exam-season spending (mean 2.06). This positions emotional regulation as the central mechanism of UPI impulse behavior in the student context. Students under academic pressure appear to use consumption as a coping mechanism, an observation supported by the NLP analysis where "sad," "stressed," and "anxiety" appeared in regret descriptions.

5.3 Food as the Dominant Impulse and Regret Category

Food delivery dominated both the word frequency analysis ("food," "order," "eat," "hungry," "aloo," "paratha" in the top words) and the regret category rankings. This is structurally explained by the near-instant delivery windows of Swiggy, Zomato, and Blinkit: the time between the impulse trigger and gratification is 20–40 minutes, providing no natural delay that might allow deliberate reconsideration.

5.4 Feature Importance and Practical Implications

The Random Forest identifies the impulse composite score (a behavioral trait, not a demographic) and transaction frequency (a usage behavior) as the two strongest predictors of financial regret. This implies that interventions should target behavioral patterns rather than demographics. Fintech applications could implement spending nudges triggered by composite trigger score thresholds or by detecting elevated transaction frequency relative to the student's historical baseline.

6 Limitations

- **Sample size:** 105 respondents is sufficient for exploratory analysis but limits the statistical power of the Random Forest classifier and may not generalize to the broader Indian college population.
- **Sample composition:** 51.4% of respondents are from IIT Bhilai, introducing institutional bias. IIT students may have different spending patterns than students at other institution types.

- **Gender imbalance:** 67.6% male respondents limits the statistical validity of gender-disaggregated findings.
- **Self-reported data:** All behavioral measures are self-reported and subject to recall bias and social desirability effects, particularly for questions about hidden purchases and running out of money.
- **NLP corpus size:** Only 21 of 105 respondents completed the open-ended question. TextBlob sentiment correlation with regret intensity ($r = 0.020$) is not statistically meaningful at this sample size. NLP findings are exploratory and descriptive only.
- **Model performance:** The Random Forest classifier achieved 62% accuracy on a 21-sample test set. Results should be interpreted as directional rather than definitive.

7 Conclusion

This study examined UPI-based impulse spending behavior and financial regret among 105 Indian college students using a multi-method approach including EDA, NLP, K-Means clustering, and Random Forest classification. Key findings are:

(1) Evening and late-night hours are the primary impulse purchase windows, with 69.5% of respondents reporting evening impulse spending.
(2) Food delivery is the most common and most regretted spending category.
(3) Stress relief and academic stress are the dominant behavioral triggers, ranking above social influence and promotional mechanics.
(4) Three spending archetypes were identified: Routine Evening Spenders (62.9%), Controlled Spenders (29.5%), and High Impulsive Spenders (7.6%), with the latter showing 100% late-night activity and the highest financial distress indicators.
(5) The impulse composite score and weekly transaction frequency are the strongest predictors of high financial regret risk.

Future work should expand the sample across more institutions and geographic regions, incorporate longitudinal tracking of spending behavior, and explore the effectiveness of digital nudge interventions targeting the identified behavioral signatures.

The full project code and processed dataset are available at: https://github.com/[username]/upi-impulse-analysis. An interactive Streamlit dashboard is deployed at: https://[username]-upi-impulse.streamlit.app.

Acknowledgments

The authors thank all 105 survey respondents for their participation. Survey responses were anonymized prior to analysis. Email addresses collected for report delivery were stored separately and never used in analytical processing.